Practice Questions for 2024-09-17-DualityGeneral

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1 Questions

1. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. If, during the process of finding the dual, we determine that $m^T - \lambda^T A - \mu^T B \neq 0$, what can we definitively conclude about the dual objective function $\xi(\lambda, \mu)$, and why?

   a. The dual objective function is bounded.

   b. The dual objective function is unbounded above.

   c. The dual objective function is unbounded below.

   d. The dual objective function has a finite minimum value.

   e. The dual objective function is equal to zero.

2. Referring to Slide 19, we have the primal linear program (P): Maximize $\xi(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y, \lambda) = c^T x - y^T(Ax - b) - \lambda^T(-x)$, where y and λ are Lagrange multipliers. Explain the significance of the non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.

   a. These constraints are arbitrary and do not affect the dual problem.

   b. These constraints ensure the Lagrangian remains bounded.

   c. These constraints are necessary to guarantee the feasibility of the dual problem.

   d. These constraints arise from the Karush-Kuhn-Tucker (KKT) conditions and are essential for optimality.

   e. These constraints are a consequence of the inequality constraints in the primal problem and are crucial for deriving the dual inequality constraints.

3. Slide 16 shows the dual objective function $\xi(\lambda, \mu)$ for a general linear program as a piecewise function. Explain the significance of the condition $\lambda^T A + \mu^T B = m^T$ in determining whether the dual objective function is bounded or unbounded.

   a. If the condition holds, the dual is unbounded; otherwise, it's bounded.

   b. If the condition holds, the dual is bounded; otherwise, it's unbounded.

   c. The condition is irrelevant to the boundedness of the dual.

   d. The condition only affects the feasibility of the dual, not its boundedness.

   e. The condition determines the optimal value of the dual, but not its boundedness.

4. Consider the primal problem (P) and dual problem (D) presented in Slide 18. Explain the relationship between the optimal values of the primal and dual problems, and state the conditions under which strong duality holds.

   a. The optimal value of (P) is always greater than the optimal value of (D).

   b. The optimal value of (P) is always less than the optimal value of (D).

   c. The optimal values of (P) and (D) are always equal.

   d. The optimal value of (P) is less than or equal to the optimal value of (D), with equality holding if both problems are feasible.

   e. The optimal value of (P) is less than or equal to the optimal value of (D), with equality holding if both problems are feasible and at least one has a finite optimal solution.

5. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. Explain how the non-negativity constraint $\lambda \geq 0$ arises in the context of the Lagrangian and its impact on the derivation of the dual problem.

   a. The constraint $\lambda \geq 0$ is arbitrarily imposed to ensure the dual problem is well-defined.

   b. The constraint $\lambda \geq 0$ arises from the Karush-Kuhn-Tucker (KKT) conditions, which are necessary for optimality in constrained optimization problems. It ensures that the inequality constraints are satisfied at the optimal solution.

   c. The constraint $\lambda \geq 0$ is a consequence of the fact that the primal problem is a maximization problem. It is needed to maintain consistency between the primal and dual problems.

   d. The constraint $\lambda \geq 0$ is a direct result of the non-negativity constraints on the primal variables x. It is necessary to ensure that the dual problem is feasible.

   e. The constraint $\lambda \geq 0$ is not strictly necessary for the derivation of the dual problem, but it simplifies the analysis and ensures that the dual objective function is bounded below.

6. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. If, during the process of finding the dual, we determine that $m^T - \lambda^T A - \mu^T B \neq 0$, what can we definitively conclude about the dual objective function $\xi(\lambda, \mu)$, and why?

   a. The dual objective function $\xi(\lambda, \mu)$ is bounded.

   b. The dual objective function $\xi(\lambda, \mu)$ is unbounded below.

   c. The dual objective function $\xi(\lambda, \mu)$ is unbounded above.

   d. The dual objective function $\xi(\lambda, \mu)$ is equal to zero.

   e. We cannot conclude anything about the dual objective function without further information.

7. Referring to Slide 19, we have the primal linear program (P): Maximize $\xi(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y, \lambda) = c^T x - y^T(Ax - b) - \lambda^T(-x)$, where y and λ are Lagrange multipliers. Explain the significance of the non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.

   a. The constraints are arbitrary and do not affect the dual problem.

   b. The constraints ensure the Lagrangian is always non-negative.

c. The constraints are necessary to guarantee the existence of a dual problem.

d. The constraints ensure the dual problem is a minimization problem.

e. The constraints arise from the non-negativity of the slack variables in the primal problem and are crucial for deriving the dual inequalities.

8. Slide 16 shows the dual objective function $\xi(\lambda, \mu)$ for a general linear program as a piecewise function. Explain the significance of the condition $\lambda^T A + \mu^T B = m^T$ in determining whether the dual objective function is bounded or unbounded.

   a. The condition is irrelevant; the dual objective function is always bounded.

   b. If the condition holds, the dual is unbounded; otherwise, it's bounded.

   c. If the condition holds, the dual is bounded; otherwise, it's unbounded.

   d. The condition determines the optimal value of the dual, not its boundedness.

   e. The condition is only relevant for primal problems with equality constraints.

9. Consider the primal problem (P) and dual problem (D) presented in Slide 18. Explain the relationship between the optimal values of the primal and dual problems, and state the conditions under which strong duality holds.

   a. The optimal value of the primal is always greater than the optimal value of the dual.

   b. The optimal values are always equal, regardless of feasibility.

   c. The optimal values are equal if both the primal and dual problems are feasible.

   d. The optimal value of the primal is always less than the optimal value of the dual.

   e. The optimal values are equal if the primal problem is feasible and bounded.

10. a. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. If, during the process of finding the dual, we determine that $m^T - \lambda^T A - \mu^T B \neq 0$, what can we definitively conclude about the dual objective function $\xi(\lambda, \mu)$, and why?

    A. The dual objective function $\xi(\lambda, \mu)$ is bounded.

    B. The dual objective function $\xi(\lambda, \mu)$ is unbounded.

    C. The dual objective function $\xi(\lambda, \mu)$ is equal to zero.

    D. The dual objective function $\xi(\lambda, \mu)$ is equal to the primal objective function.

    E. We cannot conclude anything about the dual objective function without further information.

    b. Referring to Slide 19, we have the primal linear program (P): Maximize $\xi(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y, \lambda) = c^T x - y^T(Ax - b) - \lambda^T(-x)$, where y and λ are Lagrange multipliers. Explain the significance of the non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.

    A. The non-negativity constraints are arbitrary and do not affect the dual problem.

    B. The constraints ensure that the Lagrangian is always non-negative.

    C. The constraints are necessary to guarantee the existence of a dual problem.

D. The constraints arise from the non-negativity of the slack variables in the primal problem and are crucial for deriving the dual problem's inequality constraints.

E. The constraints are needed to ensure that the dual problem is a minimization problem.

c. Slide 16 shows the dual objective function $\xi(\lambda, \mu)$ for a general linear program as a piecewise function. Explain the significance of the condition $\lambda^T A + \mu^T B = m^T$ in determining whether the dual objective function is bounded or unbounded.

A. The condition is irrelevant; the dual objective function is always bounded.

B. The condition is irrelevant; the dual objective function is always unbounded.

C. If the condition holds, the dual objective function is bounded; otherwise, it is unbounded.

D. If the condition holds, the dual objective function is unbounded; otherwise, it is bounded.

E. The condition only affects the value of the dual objective function, not whether it is bounded or unbounded.

2 Answers

1. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. If, during the process of finding the dual, we determine that $m^T - \lambda^T A - \mu^T B \neq 0$, what can we definitively conclude about the dual objective function $\xi(\lambda, \mu)$, and why?

   a. Incorrect. A non-zero coefficient for x in the Lagrangian leads to unboundedness, not boundedness.

   b. The correct answer is B. When $m^T - \lambda^T A - \mu^T B \neq 0$, the Lagrangian $L(x, \lambda, \mu) = (m^T - \lambda^T A - \mu^T B)x + \lambda^T b + \mu^T c$ is a linear function of x with a non-zero coefficient for x. Since x is unconstrained in the maximization of the Lagrangian (Slide 14, Slide 15, Slide 16), this linear function can be made arbitrarily large by choosing appropriate values of x. Therefore, the maximum of the Lagrangian, which defines the dual objective function $\xi(\lambda, \mu)$, is unbounded above.

   c. Incorrect. The Lagrangian is unbounded above, not below, in this case.

   d. Incorrect. An unbounded Lagrangian implies an unbounded dual objective function.

   e. Incorrect. The value of the dual objective function is not necessarily zero; it's unbounded.

2. Referring to Slide 19, we have the primal linear program (P): Maximize $\xi(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y, \lambda) = c^T x - y^T(Ax - b) - \lambda^T(-x)$, where y and λ are Lagrange multipliers. Explain the significance of the non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.

   a. Incorrect. These constraints are not arbitrary; they are fundamental to the duality theory.

   b. Incorrect. While these constraints help in ensuring boundedness under certain conditions, their primary significance lies in the correct formulation of the dual problem.

   c. Incorrect. These constraints are necessary for the correct formulation of the dual problem, but feasibility is a broader concept related to the satisfaction of all constraints in both the primal and dual problems.

   d. Incorrect. While related to optimality conditions, the primary role of these constraints is in the correct formulation of the dual problem itself, before considering optimality conditions.

   e. The correct answer is E. The non-negativity constraints on the Lagrange multipliers ($y \geq 0$ and $\lambda \geq 0$) are directly linked to the inequality constraints in the primal problem. The inequality $Ax \leq b$ corresponds to the non-negativity of the Lagrange multipliers y, and the inequality $x \geq 0$ (or equivalently, $-x \leq 0$) corresponds to the non-negativity of the Lagrange multipliers λ. These constraints are essential because they ensure that when maximizing the Lagrangian to obtain the dual function, the resulting dual constraints are inequalities, not equalities. Without these constraints, the dual problem would not be correctly formulated.

3. Slide 16 shows the dual objective function $\xi(\lambda, \mu)$ for a general linear program as a piecewise function. Explain the significance of the condition $\lambda^T A + \mu^T B = m^T$ in determining whether the dual objective function is bounded or unbounded.

   a. Incorrect. The opposite is true: the dual is bounded when the condition holds.

   b. The correct answer is B. The condition $\lambda^T A + \mu^T B = m^T$ arises from setting the gradient of the Lagrangian with respect to x to zero. When this condition holds, the Lagrangian is independent of x, and its maximum value is simply $\lambda^T b + \mu^T c$, which is a finite value. However, if the condition does not hold, the Lagrangian is a linear function of x with a non-zero coefficient, making it unbounded above. Therefore, the dual objective function $\xi(\lambda, \mu)$ is bounded if and only if the condition $\lambda^T A + \mu^T B = m^T$ is satisfied.

c. Incorrect. The condition is crucial in determining the boundedness of the dual objective function.

d. Incorrect. The condition directly impacts the boundedness of the dual objective function, which is a key aspect of the dual problem's properties.

e. Incorrect. While the condition determines the value of the dual objective function when it is bounded, its primary significance lies in determining whether the dual is bounded or unbounded.

4. Consider the primal problem (P) and dual problem (D) presented in Slide 18. Explain the relationship between the optimal values of the primal and dual problems, and state the conditions under which strong duality holds.

   a. Incorrect. Weak duality states that the optimal value of the primal is less than or equal to the optimal value of the dual.

   b. Incorrect. Weak duality states that the optimal value of the primal is less than or equal to the optimal value of the dual.

   c. Incorrect. While strong duality implies equality, it only holds under specific conditions of feasibility and boundedness.

   d. Incorrect. The condition for strong duality requires at least one problem to have a finite optimal solution, not just feasibility of both.

   e. The correct answer is E. Weak duality always holds, meaning the optimal value of the primal problem is less than or equal to the optimal value of the dual problem. Strong duality, where the optimal values are equal, holds under certain conditions. Specifically, strong duality holds if both the primal and dual problems are feasible and at least one of them has a finite optimal solution. If one problem is infeasible, the other may be unbounded or infeasible. If both are feasible, but one is unbounded, the other must be infeasible. Therefore, the equality of optimal values (strong duality) requires both feasibility and the existence of at least one finite optimal solution.

5. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. Explain how the non-negativity constraint $\lambda \geq 0$ arises in the context of the Lagrangian and its impact on the derivation of the dual problem.

   a. The constraint $\lambda \geq 0$ is not arbitrary; it has a fundamental mathematical basis in the KKT conditions and is essential for the correct formulation of the dual problem.

   b. The constraint $\lambda \geq 0$ is a direct consequence of the KKT conditions, which are necessary conditions for optimality in nonlinear programming problems. These conditions ensure that the inequality constraints are satisfied at the optimal solution. In the context of the Lagrangian, the non-negativity of λ is crucial for ensuring that the dual problem is a minimization problem and that the weak duality theorem holds. Without this constraint, the dual problem might be unbounded or infeasible.

   c. While the primal problem being a maximization problem influences the structure of the dual problem, the constraint $\lambda \geq 0$ is not solely a consequence of this fact. It stems from the KKT conditions and the nature of inequality constraints.

   d. The constraint $\lambda \geq 0$ is not directly related to the non-negativity constraints on the primal variables x. It arises from the KKT conditions applied to the inequality constraints $Ax \leq b$.

e. The constraint $\lambda \geq 0$ is absolutely necessary for the correct derivation and interpretation of the dual problem. Without it, the dual problem would not be a proper minimization problem, and the duality theorems would not hold.

6. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. If, during the process of finding the dual, we determine that $m^T - \lambda^T A - \mu^T B \neq 0$, what can we definitively conclude about the dual objective function $\xi(\lambda, \mu)$, and why?

a. Incorrect. If $m^T - \lambda^T A - \mu^T B \neq 0$, the Lagrangian is unbounded, meaning the dual objective function is also unbounded.

b. Incorrect. The Lagrangian is unbounded above, not below, in this case, leading to an unbounded above dual objective function.

c. The correct answer is C. The Lagrangian is given by $L(x, \lambda, \mu) = m^T x - \lambda^T (Ax - b) - \mu^T (Bx - c) = (m^T - \lambda^T A - \mu^T B)x + \lambda^T b + \mu^T c$. If $m^T - \lambda^T A - \mu^T B \neq 0$, then the Lagrangian is unbounded above as a function of x because x is unconstrained. Since $\xi(\lambda, \mu) = \max_x L(x, \lambda, \mu)$, the dual objective function is unbounded above.

d. Incorrect. The value of the dual objective function depends on the values of λ and μ, and it is unbounded in this case.

e. Incorrect. The condition $m^T - \lambda^T A - \mu^T B \neq 0$ directly implies that the dual objective function is unbounded.

7. Referring to Slide 19, we have the primal linear program (P): Maximize $\xi(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y, \lambda) = c^T x - y^T (Ax - b) - \lambda^T (-x)$, where y and λ are Lagrange multipliers. Explain the significance of the non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.

a. Incorrect. The constraints are not arbitrary; they are essential for the correct formulation of the dual problem.

b. Incorrect. The constraints ensure that the penalties for violating the primal constraints are non-negative, but they do not guarantee that the Lagrangian itself is always non-negative.

c. Incorrect. While the constraints are necessary for a meaningful dual problem, they don't guarantee its existence; the existence depends on the primal problem's properties.

d. Incorrect. The constraints on y and λ are necessary for the correct formulation of the dual inequalities, but the dual problem's minimization nature arises from the minimization of the maximum of the Lagrangian.

e. The non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ are crucial for deriving the dual problem. The Lagrangian is $L(x, y, \lambda) = c^T x - y^T (Ax - b) - \lambda^T (-x)$. When maximizing $L(x, y, \lambda)$ with respect to x, the terms $-y^T (Ax - b)$ and $-\lambda^T (-x)$ represent the penalties for violating the constraints $Ax \leq b$ and $x \geq 0$, respectively. The non-negativity of y and λ ensures that these penalties are always non-negative, leading to the correct dual inequalities. Without these constraints, the dual problem would not be correctly formulated.

8. Slide 16 shows the dual objective function $\xi(\lambda, \mu)$ for a general linear program as a piecewise function. Explain the significance of the condition $\lambda^T A + \mu^T B = m^T$ in determining whether the dual objective function is bounded or unbounded.

a. Incorrect. The condition is critical in determining the boundedness of the dual objective function.

b. Incorrect. The opposite is true: if the condition holds, the dual is bounded; otherwise, it's unbounded.

c. The condition $\lambda^T A + \mu^T B = m^T$ is crucial. The Lagrangian is $L(x, \lambda, \mu) = (m^T - \lambda^T A - \mu^T B)x + \lambda^T b + \mu^T c$. If $\lambda^T A + \mu^T B = m^T$, the term multiplying x vanishes, and the Lagrangian is constant with respect to x, resulting in a bounded dual objective function $\xi(\lambda, \mu) = \lambda^T b + \mu^T c$. However, if $\lambda^T A + \mu^T B \neq m^T$, the Lagrangian is unbounded in x, making the dual objective function unbounded.

d. Incorrect. The condition determines whether the dual is bounded or unbounded, not just its optimal value (when bounded).

e. Incorrect. The condition applies to general linear programs with both inequality and equality constraints.

9. Consider the primal problem (P) and dual problem (D) presented in Slide 18. Explain the relationship between the optimal values of the primal and dual problems, and state the conditions under which strong duality holds.

   a. Incorrect. Weak duality states that the optimal value of the primal is less than or equal to the optimal value of the dual.

   b. Incorrect. Feasibility of both problems is a necessary condition for strong duality to hold.

   c. Strong duality states that if both the primal (P) and dual (D) problems are feasible, then their optimal values are equal. Weak duality always holds ($\xi^* \leq \xi^*$), but strong duality only holds under the condition of primal and dual feasibility. If either problem is infeasible, strong duality does not hold.

   d. Incorrect. Weak duality states that the optimal value of the primal is less than or equal to the optimal value of the dual.

   e. Incorrect. While primal feasibility and boundedness are sufficient conditions for the existence of an optimal solution, dual feasibility is also required for strong duality.

10. a. Consider the general linear program (GLP) presented in Slide 11: Maximize $\xi(x) = m^T x$ subject to $Ax \leq b$ and $Bx = c$. The Lagrangian is formed, incorporating Lagrange multipliers λ and μ. If, during the process of finding the dual, we determine that $m^T - \lambda^T A - \mu^T B \neq 0$, what can we definitively conclude about the dual objective function $\xi(\lambda, \mu)$, and why?

    A. Incorrect. The condition $m^T - \lambda^T A - \mu^T B \neq 0$ implies that the Lagrangian is unbounded in x, leading to an unbounded dual objective function.

    B. The correct answer is B. If $m^T - \lambda^T A - \mu^T B \neq 0$, then the Lagrangian $L(x, \lambda, \mu) = (m^T - \lambda^T A - \mu^T B)x + \lambda^T b + \mu^T c$ is an unbounded function of x because it contains a term linear in x with a non-zero coefficient. Since $\xi(\lambda, \mu) = \max_x L(x, \lambda, \mu)$, the dual objective function is unbounded.

    C. Incorrect. The value of the dual objective function depends on the values of λ and μ, and the condition given does not imply that it is zero.

    D. Incorrect. The primal and dual objective functions are only equal under specific conditions (strong duality), and this condition alone does not guarantee equality.

    E. Incorrect. The condition $m^T - \lambda^T A - \mu^T B \neq 0$ directly implies the unboundedness of the dual objective function.

    b. Referring to Slide 19, we have the primal linear program (P): Maximize $\xi(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y, \lambda) = c^T x - y^T(Ax - b) - \lambda^T(-x)$, where y

and λ are Lagrange multipliers. Explain the significance of the non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.

A. Incorrect. The non-negativity constraints are fundamental to the derivation of the dual problem and are not arbitrary.

B. Incorrect. The Lagrangian itself can be negative; the non-negativity constraints are on the Lagrange multipliers.

C. Incorrect. While the constraints are crucial, they don't guarantee the existence of a dual problem; the existence is guaranteed under broader conditions.

D. The correct answer is D. The non-negativity constraints $y \geq 0$ and $\lambda \geq 0$ are not arbitrary. They arise directly from the nature of the inequality constraints in the primal problem. The Lagrange multipliers y correspond to the constraints $Ax \leq b$, and the multipliers λ correspond to the constraints $x \geq 0$ (equivalently, $-x \leq 0$). The non-negativity of these multipliers is essential for ensuring that the Lagrangian correctly reflects the constraints of the primal problem and for deriving the inequality constraints in the dual problem. Without these constraints, the derivation of the dual problem would be incorrect.

E. Incorrect. The nature of the dual problem (minimization) arises from the primal problem being a maximization problem and the Lagrangian's structure, not directly from the non-negativity constraints on the multipliers.

c. Slide 16 shows the dual objective function $\xi(\lambda, \mu)$ for a general linear program as a piecewise function. Explain the significance of the condition $\lambda^T A + \mu^T B = m^T$ in determining whether the dual objective function is bounded or unbounded.

A. Incorrect. The condition is directly related to the boundedness of the dual objective function.

B. Incorrect. The dual objective function is bounded when the condition holds.

C. The correct answer is C. The condition $\lambda^T A + \mu^T B = m^T$ is crucial. When this condition is met, the term involving x in the Lagrangian vanishes, resulting in a constant value for the Lagrangian, and thus a bounded dual objective function $\xi(\lambda, \mu) = \lambda^T b + \mu^T c$. However, if $\lambda^T A + \mu^T B \neq m^T$, the Lagrangian contains a term linear in x with a non-zero coefficient, making it unbounded. Therefore, the dual objective function $\xi(\lambda, \mu)$ is unbounded in this case.

D. Incorrect. The dual objective function is unbounded when the condition does not hold.

E. Incorrect. The condition determines whether the dual objective function is bounded or unbounded.